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Juvenile Crime and Anticipated Punishment[†]

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Can sanctions deter juvenile crime? Research indicates that they may not, as offending barely decreases when individuals cross the age of criminal majority and begin to face harsher sanctions. Several models of criminal behavior predict, however, that these small reactions close to the threshold may mask larger behavioral responses among individuals below the age threshold. Policy variation between 2007–2015 in the United States is used to show evidence consistent with these predictions—juvenile crime increases when the age of majority is increased. This increase is driven by younger age groups and is considerably larger than discontinuity estimates at the threshold. (*JEL* D91, J13, K42)

Recent research in economics and criminology indicates that the threat of punitive sanctions may not discourage youth from committing crime.[1] This understanding is reflected in the public policy shift toward increasing rehabilitation efforts and reducing punitive sanctions for juveniles across the United States. One example of this shift is the growing popularity of raise-the-age reforms, which increase the age of criminal majority—the youngest age at which individuals are processed in the adult criminal justice system within a state—in order to increase the chances that juveniles and young adults are processed within a less punitive system.

Qualitative evidence, however, does not support the view that changes in sanctions at the age of majority do not discourage crime. Youth involved in crime report consciously desisting from criminal activity close to the age of criminal majority, driven by the differences they perceive in the treatment meted out by juvenile and adult criminal justice systems (Glassner et al. 1983; Hekman, Gold, and Ruhland 1983; Fagan and Piquero 2007). Law enforcement officials also voice concerns about the

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[1] Two strands of the literature examine the impact of sanctions on crime. The first studies adult offenders and finds evidence of a deterrent effect of sanction severity on crime (Helland and Tabarrok 2007; Drago, Galbiati, and Vertova 2009; Bell, Jaitman, and Machin 2014; Hansen 2015). The second uses regression discontinuity designs to examine the impact of differences in sentence severity between juvenile and adult offenders, and it finds little evidence of deterrence among young offenders (Hjalmarsson 2009a; Lee and McCrary 2017), though this finding has been contested by recent work (Lovett and Xue 2018). This paper ties in with the second set of papers and speaks to deterrence effects among young offenders.

potential for heightened recruitment of juveniles into gangs and ensuing violence in response to raising the age of criminal majority.[2] While this divergence may be driven by methodological differences, it may also be explained by the fact that the empirical literature has attempted to identify deterrence effects among adolescents by looking for discontinuous decreases in offending when an individual reaches the age of majority and faces more severe sanctions. Underlying this approach is an understanding of adolescent crime as a series of on-the-spot decisions, with no dependence on, say, previous criminal involvement or the criminal involvement of others. These models predict, however, that the discontinuity at the age of majority may be smaller and much more difficult to detect than in the absence of these forces.[3] This paper uses the insights of these alternative models of criminal behavior to provide novel evidence on the effect of sanctions on juvenile crime.

Several models of criminal behavior may predict a behavioral response among individuals below the age of majority. To anchor the discussion, I focus on two such models that are grounded in empirical support. The first is a dynamic model in which individuals evaluate the costs and benefits of crime in each period, but also accumulate criminal experience as they commit crime, which increases the return to committing crime in the future.[4] The second is a static model under which individuals evaluate the costs and benefits of crime in each period, but the return to crime is influenced by their peers' decisions to commit crime.[5] When the age of criminal majority is raised from 17 to 18, both models predict that individuals *younger* than 17 increase criminal activity, not just 17-year-olds.

Before turning to the empirical strategy and causal estimates, I use *self-reported* data on criminal involvement to conduct a first litmus test. Data from the 1997 National Longitudinal Survey of Youth is used to document patterns of criminal involvement by age, separating states by their age of criminal majority (henceforth, ACM). Consistent with the predictions of these models, individuals in states that set the ACM at 18 are about 1 percentage point more likely to have committed a crime within the last 12 months, an 8 percent increase relative to the mean. Further, this difference is driven entirely by juveniles and young adults (ages 13–21), not by older adults.

To identify the causal impact of the ACM increase on crime, I use state-specific policy changes in six contiguous states in the Northeast—Connecticut, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont. Since the mid-2000s, each

[2] For instance, see home.chicagopolice.org/community/gang-awareness and www.dnainfo.com/new-york/raise-the-age.

[3] An additional issue is that crime may be underrecorded at higher rates for juveniles (those below the age of criminal majority) than adults. As a result, discontinuity estimates at the age of criminal majority may show that sanctions have negligible or even positive effects on crime. Among others, Greenwood (1995); Chalfin and McCrary (2014); Loeffler and Chalfin (2017); and Costa et al. (2018) discuss concerns about differential underrecording for juvenile and adult crimes. The difference is likely driven by the fact that law enforcement officials must meet additional requirements to arrest and hold juveniles in custody.

[4] This framework is motivated by research that shows exposure to crime and criminal experience can increase the return to future offending; this includes Bayer et al. (2009); Pyrooz, Sweeten, and Piquero (2013); Damm and Dustmann (2014); Carvalho and Soares (2016); Stevenson (2017); and Damm and Gorinas (2020).

[5] This framework is motivated by research that shows juvenile crime is affected by peer influences within schools and neighborhoods; this includes Kline (2011); Drago and Galbiati (2012); Patacchini and Zenou (2012); Billings, Deming, and Ross (2019); and Billings and Schnepel (2020).

of these states has successfully passed raise-the-age reforms, lending credibility to the assumption that the actual timing of legislation passage was unrelated to local crime trends.[6] Causal estimates are generated via a difference-in-difference-in-differences framework, which leverages variation in the effects of the policy across age groups, states, and time; specifically, the empirical strategy compares those who are predicted to be affected by the ACM increase (adolescents close to the ACM) with individuals that are not (those further from the ACM, adults older than 21) across treatment and comparison states, before and after the ACM changes.[7]

Regression estimates are based on a balanced panel of law enforcement agencies that report monthly arrests by age during the period of 2005–2016 to the FBI. To minimize underreporting concerns to the extent possible, attention is restricted to offenses that are regarded as serious and tend to be reported more reliably than other crime categories—i.e., the core Index 1 violent (murder, robbery, and aggravated assault) and property (burglary, larceny, and motor vehicle theft) crimes.[8] Prior to the ACM increase from 17 to 18, there is no statistical difference between the treatment and comparison group coefficients. When the ACM is raised, arrest rates for 13–21-year-olds increase by 12 percent of the mean, an effect that is primarily driven by 13–16-year-olds.[9] These estimates are meaningfully larger than discontinuity estimates documented at the threshold—for instance, arrest rates for 16-year-olds (the age group closest to the threshold) rise by 34–37 percent when the ACM increases; discontinuity estimates, on the other hand, have ranged from negligible to 12 percent (Lee and McCrary 2017; Lovett and Xue 2018).

While these estimates show that raising the ACM is not costless, these findings should not be taken as an unequivocal rejection of raise-the-age reforms. Rather, they highlight the need for more research to quantify the benefits of these policies, which are not well understood. For instance, proponents of raising the ACM usually argue that being processed as a juvenile will lower recidivism, which is not supported by recent studies (Entorf 2011; Pichler and Römer 2013; Loeffler and Grunwald 2015). Understanding both the costs and benefits of these policies is of particular relevance today, as states have continued to raise their ACMs, sometimes well above 18—Vermont, for instance, has raised the ACM to 21 for some offenses, while states like Connecticut, Illinois, and Massachusetts have introduced legislation to do the same.[10]

This study makes several contributions to the academic literature on sanctions and juvenile crime: (1) it evaluates the impact of raise-the-age reforms, which have

[6] Estimates are robust to including all states in the United States; these tests are discussed in Section VC.

[7] Recent work on difference-in-differences(-in-differences) with staggered treatment dates has identified potential issues with conventional two-way (three-way) fixed effects estimators. Section VA shows that the main results are only strengthened when estimated using Borusyak, Jaravel, and Spiess (2021)'s robust and efficient estimator that circumvents these issues.

[8] Rape is excluded because the FBI definition for this offense changed in 2013, which is in middle of the study period.

[9] Unlike regression discontinuity estimates at the threshold, estimates based on younger cohorts are unconfounded by other changes at the age of majority. Among others, these may include changes in underrecording (i.e., juvenile crime may be recorded at lower rates than adult crime), incapacitation effects (convicted juveniles face shorter sentences than convicted adults), aging out of foster care, and aging out of compulsory school attendance.

[10] For more details, see https://www.bostonglobe.com/metro/2019/07/09/crime-bill-would-redefine-juveniles-age/maHshbBT6QaaX9ooVDVidN/story.html.

become an increasingly popular policy in the United States. (2) It offers an explanation for why previous research on sanctions at the age of majority and juvenile crime may have found contradictory results—Hjalmarsson (2009a); Costa et al. (2018); Lee and McCrary (2017); Loeffler and Chalfin (2017); and Damm et al. (2017) fail to find an effect, while Levitt (1998), Oka (2009), Munyo (2015), and Lovett and Xue (2018) do.[11] Loeffler and Chalfin (2017), for instance, study the impact of raising the ACM on juvenile crime but use cohorts *below* the original ACM as the comparison group—if younger cohorts also respond to the ACM increase, as is shown in the current paper, then this approach would bias their estimates toward zero. (3) It shows that sanctions may have heterogeneous effects across race groups—the estimates discussed above are largely driven by an increase in the arrest rate for White youth, while the response among Black youth is smaller in magnitude and statistically indistinguishable from zero. This is consistent with effective treatment differing across race groups—if Black youth are more likely to be charged in adult courts (Juszkiewicz 2009; Zane, Cochran, and Mears 2022) or face harsher sanctions within the juvenile justice system (Eren and Mocan 2018), raising the ACM may change their incentives less than those of other youth.[12] (4) It contributes to research on the impact of sanctions on crime in general, which has also found mixed results (Nagin 2013; Chalfin and McCrary 2014; O'Flaherty and Sethi 2014).

The rest of this paper is organized into six sections. Section I describes trends in juvenile crime in the United States and states' policy responses to these trends since the 1990s. Section II describes the datasets used in this paper and shows that crime committed by juveniles is likely to be recorded by law enforcement agencies at lower rates than crime by adults. Section III describes two models of criminal behavior and their predictions on how adolescent crime will respond to an increase in the ACM. Section IV describes the difference-in-difference-in-differences strategy, and Section V exploits policy variation in the ACM across time and space in the United States to estimate the causal impact of the ACM increase on arrest rates. Section VI concludes.

I. Setting

This section provides a brief description of juvenile crime trends in the United States and policy responses to these trends. Changes to the ACM in states across the United States are described at some length, as they are used to estimate the causal impact of sanctions on juvenile crime in later sections. Finally, I use *self-reported*

[11] Damm et al. (2017) study effects at the age of criminal *responsibility*, the age at which individuals are transferred from the social service system to the juvenile justice system; most papers focus on the age of criminal *majority*, the age at which individuals are transferred from the juvenile to the adult criminal justice system. Individuals between the ages of criminal *responsibility* and *majority* in Denmark benefit from a number of sentencing policies and options not available for adults (Kyvsgaard 2004), which makes the comparison to the US setting inexact. Costa et al. (2018) use homicide victimization rates by age as a proxy for criminal activity (as opposed to arrests or offenses). Oka (2009) also finds that harsher punishments for 14–15-year-olds in Japan reduced criminal offending by 13-year-olds.

[12] This finding is supported by a recent study on the effect of sanctions on juvenile crime in California—using a regression discontinuity design, Lovett and Xue (2018) find that White and Asian youth are responsive to the sanction hike at the ACM in California, while Black youth are not.

data on criminal involvement to show that adolescent crime is higher in states that set the ACM at 18 as opposed to 17. While not causal, this evidence is intended to motivate the theoretical discussion and empirical analysis that follows.[13]

A. *Juvenile Crime: Trends and Policy Responses*

The roots of the juvenile justice system in the United States can be traced back to the nineteenth century, when the desire to remove juveniles from overcrowded adult prisons led to the development of separate facilities for juveniles, as well as options like out-of-home placement and probation. The juvenile justice system in the United States today also includes a separate system of juvenile courts, where decisions are made by an individual judge instead of a jury. Further, sentence lengths are shorter, and living conditions are generally better in juvenile incarceration facilities (Myers 2003, Lee and McCrary 2017). Additionally, it is easier to expunge or seal one's criminal records if the offense was committed as a juvenile (Litwok 2014).

States vary in who they treat as juveniles under their criminal justice systems. The ACM—the lowest age at which offenders are automatically processed as adults in the criminal justice system—has varied considerably across time and space within the United States.[14] Online Appendix Table A.1 displays a complete list of states and their ACM in 2019, and whether states have had a different ACM in the past. While the majority of states set the ACM at 17 or 18, the ACM has varied from 19 in Wyoming in 1993 to 16 in Connecticut, New York, and North Carolina in the 2010s. In the last few years, states have begun to experiment with even more ambitious legislation; Vermont has raised the ACM to 21 for some offenses, and states like Connecticut, Illinois, and Massachusetts have proposed bills along the same lines.[15]

Trends in juvenile crime explain some of the variation in the ACM over time. Online Appendix Figure A.1 plots juvenile and adult arrest rates in the United States for the period 1980–2012. Following the sharp increase in juvenile arrest rates in the 1990s (a trend that was not mirrored by adult arrest rates), states began to "get tough" on juvenile crime, passing laws that increased the severity of juvenile sanctions. Between 1992 and 1997, all but three states passed legislation easing the transfer of juveniles into the adult system, instituted mandatory minimum sentences for serious offenses, reduced juvenile record confidentiality, increased victim rights, or simply lowered the ACM (Snyder and Sickmund 2006). As shown in online Appendix Table A.1, New Hampshire, Wisconsin, and Wyoming lowered their ACMs during this period. However, the simultaneous enactment of policy changes in other states makes it unlikely that a difference-in-differences analysis would isolate the causal effect of the ACM on crime. The empirical analysis, therefore, relies on more recent

[13] All individuals in the sample of self-reported data are aged 21 or older by December 31, 2005. Since there were no ACM changes during the period 1997–2005 (see online Appendix Table A.1 for more details on ACM changes in the United States), it was not possible to conduct a difference-in-difference analysis leveraging policy variation in state-specific ACMs using this data.

[14] Some states have statutory exclusion laws in place, which allow offenders younger than this threshold to be tried as adults for serious offenses like murder.

[15] For more details, see https://www.bostonglobe.com/metro/2019/07/09/crime-bill-would-redefine-juveniles-age/maHshbBT6QaaX9ooVDVidN/story.html.

changes in states' ACMs where the identification assumptions are more likely to hold. These changes are described in detail below.

B. *ACM Changes in the 2000s*

Juvenile crime rates have fallen consistently since the 1990s, generating public support for easing juvenile sanctions and raising the ACM in states that set it below 18. ACM changes have also been catalyzed by the passage of the 2003 Prison Rape Elimination Act (PREA), a federal law aimed at preventing sexual assault in prison facilities. The PREA went into effect in 2018, and requires offenders under 18 to be housed separately from adults in correctional facilities, *irrespective* of the state's ACM. Naturally, this requirement is harder to meet in states that incarcerate 16- and 17-year-olds along with older inmates in adult facilities, and most "raise-the-age" reforms have taken place in states with ACMs below 18.[16]

Since 2003 several states have passed legislation to raise their ACM to 18.[17] The empirical analysis uses ACM changes in six contiguous states in the Northeast—Connecticut, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont. These states provide an arguably ideal setting in which to study the impact of ACM changes, for two distinct reasons. First, each of these states has introduced legislation to change the ACM since the passage of the PREA, and each has been successful. This lends credibility to the assumption that the actual timing of legislation passage was unrelated to local crime trends. Second, their geographical proximity makes it likely that unobserved factors evolved similarly across states. For these reasons, the empirical analysis focuses on a balanced sample of jurisdictions within these six states for the period 2005–2016.

C. *Self-Reported Criminal Involvement*

To motivate the theoretical framework and empirical analysis, I examine patterns of criminal involvement by age using self-reported panel data. Data comes from the National Longitudinal Survey of Youth, described in more detail in Section II. Starting in 1997, approximately 9,000 youths were asked about criminal involvement (property, drug, assault and theft offenses) in the 12 months preceding an annual interview. These self-reports show that criminal involvement is lower in states that set the ACM at 17 instead of 18, and that this difference is driven by adolescents (i.e., those aged 21 or below).

Table 1 reports results from a regression of an indicator for self-reported criminal involvement on whether the ACM is 18, controlling for the individual's sex and race. Since the regression uses a panel dataset that tracks the same individuals over time, standard errors are clustered at the individual level. Individuals in states that set the ACM at 18 are 1 percentage point more likely to have committed a crime within the

[16] A state's ACM is usually an artefact of the time period in which it established its juvenile justice system. For instance, New York set its ACM at 16 in 1909, while other states settled upon higher ACMs over the ensuing decades.

[17] For more details, see online Appendix Table A.1.

Table 1—Self-Reported Criminal Involvement

	All age groups	Age $\leq$ 21	Age $>$ 21
$\mathbf{1}\{\textit{Age of criminal majority} = 18\}$	0.008	0.013	0.002
	(0.003)	(0.005)	(0.003)
Female	−0.070	−0.103	−0.028
	(0.003)	(0.005)	(0.003)
Black	0.00002	0.0004	0.003
	(0.004)	(0.006)	(0.003)
Hispanic	−0.005	−0.008	0.001
	(0.004)	(0.006)	(0.003)
Constant	0.136	0.207	0.044
	(0.004)	(0.006)	(0.003)
Mean	0.105	0.163	0.032
Observations	96,024	53,457	42,567
Clusters	8,268	8,062	7,641

Notes: This table displays results from OLS regressions of self-reported criminal involvement on $\mathbf{1}\{\textit{Age of Criminal Majority} = 18\}$, sex, and race. The sample includes individuals in states that set the ACM at 17 or 18. Standard errors are clustered at the individual level.

Source: 1997–2015 data from the National Longitudinal Survey of Youth

last 12 months, a 7.6 percent increase relative to the mean. Separating the sample into adolescent (age 21 and below) and older (ages 22 and above) individuals, we see that this relationship is driven by heightened criminal involvement among adolescents, while there is no statistically significant difference for adults across the two groups of states.

Figure 1 examines whether this relationship between criminal involvement and the ACM begins even before individuals have reached the ACM. Coefficients from a regression of criminal involvement on age fixed effects for states that set the ACM at 17 are marked in gray, while analogous estimates for individuals in states that set the ACM at 18 are marked in black. Irrespective of the ACM, there is a clear upward trend until 14, and a steady decline thereafter.

Readers familiar with age-crime curves may notice that the pattern depicted here is different from that reported by other sources in that the peak is at 14 rather than at around 18. The main difference is not the data used but the variable plotted—here, the focus is on *any* criminal involvement by age, and is consistent with studies that look at the extensive margin, such as Pyrooz and Sweeten (2015); arrest rates by age, as in Figure 2, tend to peak at around 18, but are influenced by both intensive and extensive margin decisions to commit crime.[18] The most notable feature of this graph, however, is that the coefficients are higher at *all* ages in states that set the ACM at 18. In Section III, I argue that this pattern is consistent with several models of criminal behavior, including criminal capital accumulation and peer effects. The empirical analysis then proceeds to show that this relationship is causal—raising the ACM increases criminal activity among age groups close to the threshold.

[18] Online Appendix Figure A.2 depicts self-reported criminal involvement after restricting attention to those who were surveyed at least once at age 12, 13, or 14. In each of these subsamples, the decreasing pattern at/after age 14 is maintained.

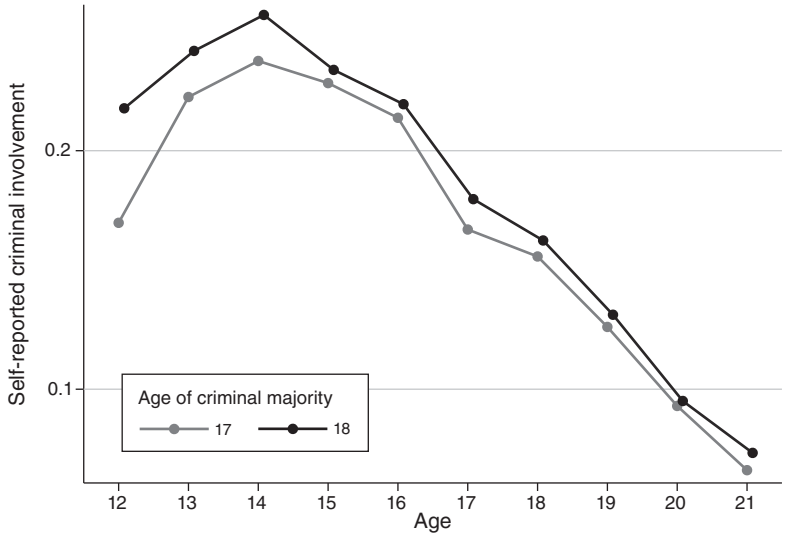

FIGURE 1. SELF-REPORTED CRIMINAL INVOLVEMENT BY AGE

Notes: This graph displays results from OLS regressions of self-reported criminal involvement on age fixed effects, separating states by their ACM; the black line displays coefficients for states that set the ACM at 18, and the gray line displays coefficients for states that set the ACM at 17.

Source: 1997–2015 data from the National Longitudinal Survey of Youth

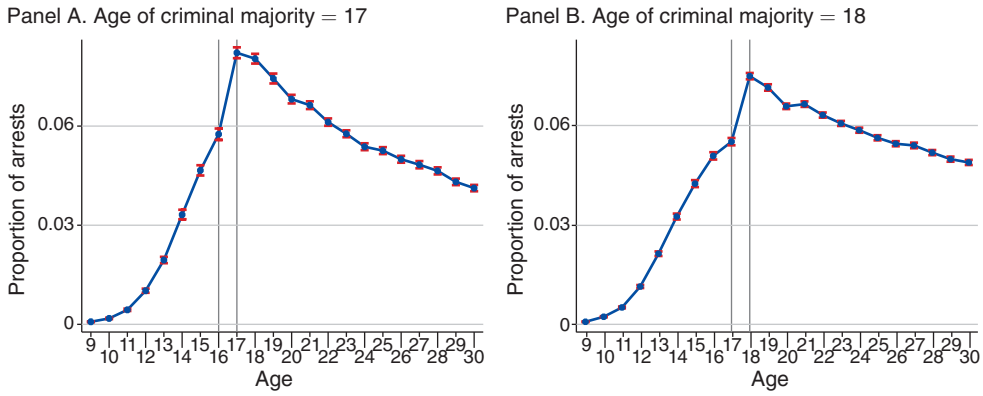

FIGURE 2. ARRESTS RECORDED BY POLICE INCREASE AT THE AGE OF CRIMINAL MAJORITY

Notes: These graphs display results from OLS regressions of the proportion of arrests for six Index crimes (homicide, assault, robbery, burglary, larceny, and motor vehicle theft) at the law enforcement agency level on age fixed effects. Standard errors are clustered at the agency level, and 95 percent confidence intervals are marked in red. Online Appendix Figures A.3 and A.4 present analogous results for Index and non-Index offenses known to the police (irrespective of arrest).

Source: 2006–2014 data from the National Incident-Based Reporting System

II. Data and Underrecording

This section describes the datasets used in the empirical analysis, and presents evidence that crimes by those below the ACM are less likely to be recorded by law enforcement officials than crimes by those above. The empirical analysis accounts for this recording issue by focusing on cohorts that are below the ACM before and after the policy change—i.e., cohorts for which the degree of underrecording is unlikely to change.

A. *Data Sources*

Local law enforcement agencies in the United States report crime statistics to federal agencies through the Uniform Crime Reports (UCR) or the National Incident Based Reporting System (NIBRS); the former covers more law enforcement agencies in the United States, while the latter presents a more detailed picture of crime within the agencies that it covers. The empirical analysis uses both of these data sources. Throughout the paper I use "recorded crime" to refer to UCR and NIBRS data, which captures the fact that researchers can only observe arrests and/or offenses that are recorded by law enforcement officials.

The UCR arrest data contain monthly counts of criminal activity within an agency's jurisdiction, with subtotals by arrestee age and sex under each offense category (FBI 2005–2016). As of 2015, law enforcement agencies representing over 90 percent of the US population submitted their crime data via the UCR. The main estimates in this paper are based on a balanced sample of agencies reporting crime via the UCR for the period 2005–2016.[19]

The NIBRS collects detailed information on each crime occurrence known to the police, and generates data as a by-product of automated records management systems (FBI 2006–2014). Crucially, offender profiles are generated independent of arrest using victim and witness statements. This allows us to examine separately whether the recording of offenses, not just arrests, is influenced by the age of the offender. As of 2017, law enforcement agencies representing 32 percent of the population submitted their crime data via the NIBRS.

Finally, data from the National Longitudinal Survey of Youth 1997 (NLSY97), a nationally representative sample of approximately 9,000 youths who were 12 to 16 years old as of December 31, 1996, is used to analyze self-reported data on criminal involvement (BLS 1997–2015). This dataset includes self-reported involvement in property, drug, assault, and theft offenses in the 12 months preceding the survey, starting in the year 1997. I use these responses as representative of the age at which the respondent spent the majority of the previous 12 months and create age profiles for criminal involvement, separating states by their ACM in Section IC.

[19] Prior research has noted the existence of errors in the UCR data, and the need for thorough cleaning before use. Section IV describes several steps that were implemented prior to using the data in order to minimize data quality concerns.

B. *Juvenile Crime Data: Evidence of Underrecording*

Criminal activity is not directly observable, so researchers usually rely on proxies like arrest and offense data collected by local law enforcement agencies. A shared concern of papers that use such data is that many steps lie between the criminal offense and the generation of an official record (Loeffler and Chalfin 2017; Costa et al. 2018).[20] Official data cannot reflect, for instance, the amount of crime that is not reported to the police, or crime that goes unrecorded due to the discretionary practices of individual officers.[21]

This issue is particularly salient for research on the effects of age-based criminal sanctions. This is because offense and arrest records are *more* likely to be generated if offenders are treated as adults by the criminal justice system.[22] Law enforcement officials must comply with additional supervisory requirements while juveniles are held in custody—unlike adults, juveniles cannot be dropped off at the local or county jail. Additionally, juveniles can only be detained for 48 hours while charges are filed in juvenile court. These additional costs make it less likely that juveniles are officially arrested or charged and, therefore, less likely to have their offenses included in official crime statistics. This is problematic for studies that compare individuals on either side of the ACM because crime is recorded at higher rates for individuals that face *lower* incentives to commit crime (individuals above the ACM). If the drop in crime committed is offset by the increased probability of a crime being recorded, research is likely to find small or null deterrence estimates. The latter effect may even dominate the former, leading to a *rise* in recorded crime exactly when the incentives to commit crime decrease, as in Lee and McCrary (2017) and Costa et al. (2018), or a *fall* in recorded crime exactly when the incentives to commit crime increase, as in Loeffler and Chalfin (2017).

I use an analogous argument to show that underrecording falls at the ACM across the United States. Recorded crime increases sharply at age 17 in states that set the ACM at 17, while this increase appears at age 18 in states that set the ACM at 18. Using monthly data at the law enforcement agency level for the years 2006–2014, panel A of Figure 2 displays the proportion of arrests attributable to each age group in states that set the ACM at 17. Panel B repeats this exercise for states that set the ACM at 18. The spike in recorded crime is striking as we transition from the age just before the threshold (16 or 17) to the age where individuals are treated as adults by the criminal justice system (17 or 18).[23] This indicates that underrecording falls as individuals cross the ACM. Existing research that compare juveniles with adults may, therefore, have generated estimates of deterrence that were adulterated by the effects of reduced underrecording.

[20] How crime statistics are generated is also a long-standing concern in criminology—see Black (1970), Black (1971), and Smith and Visher (1981).

[21] The National Crime Victimization Surveys from 2006–2010 show that less than half of all violent victimizations are reported to the police. Moreover, crimes against victims in the age group 12 to 17 were most likely to not be reported.

[22] For additional discussion, see Loeffler and Grunwald (2015a) and Loeffler and Chalfin (2017).

[23] Online Appendix Figure A.5 uses a balanced panel of agencies reporting monthly UCR data between 2006–2014 to show that this is not a NIBRS-specific feature—the largest spike in arrest rates is at age 17 in states where the ACM is 17, and at 18 in states where the ACM is 18.

What are possible workarounds to accurately measure criminal activity? One way to circumvent this issue is to use data that is less likely to be underrecorded. For instance, Costa et al. (2018) study violent death rates around the ACM in Brazil as a proxy for involvement in violent crime, as death rates are more likely to be accurately measured than crime rates. They also highlight the main drawback of this measure—violent death rates may not be reflective of trends in other types of crime. In a similar vein, some studies on crime in the United States use data on offenses known to the police instead of arrests (Abrams 2012, Loeffler and Chalfin 2017), since the latter are more likely to be affected by police officer behavior. However, the pattern documented above for arrests holds true even when we look at offenses by age, as shown in online Appendix Figures A.3 and A.4. There is a clear spike in the proportion of offenses attributable to 18-year-olds in states that set the ACM at 18, but not in states that set the ACM at 17. This indicates that data on both arrests of and offenses by those below the ACM are less likely to be reflected in official crime statistics.

This paper proposes an alternative approach to measure the impact on criminal activity—examining responses among cohorts for whom the degree of underrecording is held fixed. I test for responses to increases in the ACM among individuals who are always treated as juveniles, i.e., those to the left of the former ACM, and those who are always treated as adults, i.e., those to the right of the new ACM. Since these age groups are treated as juveniles (adults) both before and after the ACM change, the degree of underrecording for these groups should be unchanged. If adolescents to the left (right) of the threshold increase criminal activity when the ACM is moved further away from (closer to) them, and the degree of underrecording is unchanged, recorded crime should increase as well. Before formally testing for these reactions in the data, I discuss two models of criminal behavior that predict responses among cohorts who are not directly impacted by the ACM change—i.e., those below the old ACM and those above the new ACM.

III. Models of Criminal Behavior

This section discusses two models of criminal behavior that are supported by prior research, and their predictions on how adolescent crime is likely to respond to an increase in the ACM. Both models predict responses by age groups below and above the ACM. Crucially, these models identify cohorts for which underrecording is unlikely to obfuscate the policy's effects.

A. *Criminal Capital*

This section discusses the predictions of a model in which current criminal activity increases the return to criminal activity in the future. Individuals commit a crime if the benefits outweigh the costs as in Becker (1968), but also accumulate criminal capital as they commit crimes. Criminal capital increases the return to committing crime in the future and has been conceptualized of in at least three ways by prior studies: (i) the accumulation of criminal skills, as in Bayer, Hjalmarsson, and Pozen (2009); Pyrooz, Sweeten, and Piquero (2013); Damm and Dustmann

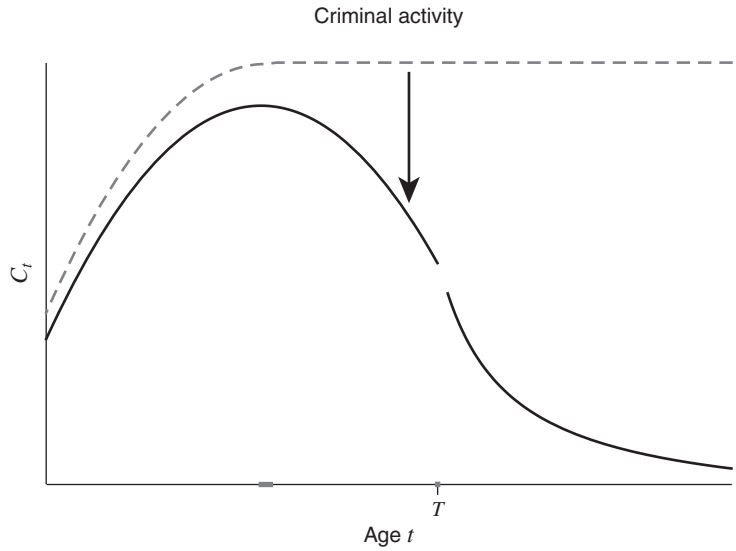

FIGURE 3. CRIMINAL ACTIVITY UNDER ANTICIPATED ADULT SANCTIONS

Notes: This figure summarizes the qualitative predictions of a model in which individuals weigh the costs and benefits of committing crimes in each period, but also accumulate criminal capital as they commit crime (which increases the return to committing crime in the future). The dashed line displays crime c at each age t if sanctions were not to increase at the age of criminal majority T. The undashed line shows that when sanctions increase at the age of criminal majority T, crime c_t is predicted to decrease discontinuously at T, but is also lower *prior* to age T. For additional details on the model, see online Appendix A.3.

(2014); Carvalho and Soares (2016); and Damm and Gorinas (2020);[24] (ii) the loss of noncriminal skills which may increase the relative return to committing crime, as in Hjalmarsson (2008); Deming (2011); and Aizer and Doyle (2015); (iii) the accumulation of noncognitive traits such as crime-oriented attitudes and behavioral habits, as in Blattman, Jamison, and Sheridan (2017); Heller et al. (2017); Stevenson (2017); and Dinarte and Egana-delSol (2019). Below, I discuss the intuition behind the model's main predictions, and relegate specifics of the theoretical framework to online Appendix A.3.[25]

As a benchmark, the dashed line in Figure 3 displays the model's predictions in the absence of harsher sanctions at an age threshold T; criminal activity c continues to increase until the steady level of criminal experience is reached, and is sustained thereafter. The anticipation of adult sanctions beginning at age T, however, generates alternative predictions, summarized by the (undashed) black line in

[24] Pyrooz, Sweeten, and Piquero (2013) and Carvalho and Soares (2016) show that embeddedness and wages in gangs increase with participation in gang-related crime. Also see Levitt and Venkatesh (2000) who find that gang members are motivated by the symbolic value attached to upward mobility in drug gangs, as well as the tournament for future riches. Criminal experience may also reduce the probability of being caught and thereby increase the net benefit of future crime.

[25] Prior research that uses a dynamic approach to study the effect of criminal sanctions includes Flinn (1986); O'Flaherty (1998); Imai and Krishna (2004); Mocan, Billups, and Overland (2005); McCrary (2010); and Munyo (2015).

Figure 3. Initially, adult sanctions entail only small deterrence effects. As individuals approach the ACM T, however, adult sanctions begin to generate larger deterrence effects: the older the age (closer to T) at which an individual commits crime, the shorter the duration for which they can benefit from its returns as a juvenile. This lowers the lifetime return to investing in criminal capital, and lowers criminal activity even *before* an individual has reached the ACM, consistent with self-reported data as seen in Figure 1. Finally, the increase in sanctions at T lowers the immediate return to criminal activity, leading to a discontinuous decrease in criminal activity at age T. The main takeaway from this figure, however, is that the increase in sanctions also leads to a decrease in crime *before* individuals have reached the ACM.

Figure 4 summarizes the model's predictions in response to an increase in the ACM from T to T'. Three groups are expected to increase criminal activity: age groups below T, age groups between T and T', and age groups above T'. In Figure 4, T and T' are marked as the age thresholds 17 and 18 (in parentheses), in reference to the empirical analysis in Section V, which estimates the impact of raising the ACM from 17 to 18.

The first group predicted to respond to the ACM increase is those below the original threshold T—when the ACM is increased, these individuals will face lower, juvenile-level sanctions for an additional year. This increases the lifetime return to committing crime, and in response, individuals below the age of T are predicted to increase criminal activity. Crucially, this response is unlikely to be offset by changes in underrecording, because those below T are treated as juveniles both before and after the policy change.

The second set of beneficiaries is those aged between T and T'. This group was subject to higher, adult-level sanctions before the policy change, but subject to lower, juvenile level sanctions after the policy change, at least until they reach age T'. This causes an increase in criminal activity through two separate channels. First, the immediate return to criminal activity is higher, because expected sanctions (if caught) in the current period are lower; second, the lifetime return to committing crime is higher, since individuals can benefit from criminal experience as juveniles for up to a year. However, this group is also subject to a simultaneous increase in underrecording since they were treated as adults previously, but are now treated as juveniles by the criminal justice system. Therefore, we may not observe an increase in official crime statistics for age groups between T and T', even if the decrease in sanctions leads to an increase in criminal activity.

Finally, an ACM increase from T to T' will also lead to a *delayed* increase in criminal activity at age T' and above. Those who are already aged T' or older when the ACM is increased experience no changes to their cost-benefit calculations of committing crime, and are not predicted to change their level of criminal activity. However, individuals will now *enter* age T' with higher levels of criminal capital than under the ACM T, raising their return to committing crime. Crucially, these age groups are treated as adults both before and after the ACM change, and the predicted increase in criminal activity is unlikely to be offset by changes in underrecording.

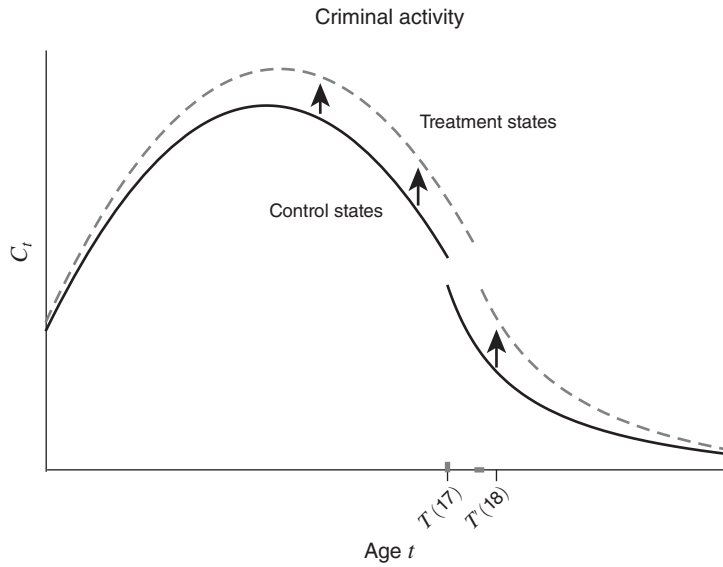

FIGURE 4. CRIMINAL ACTIVITY UNDER DELAYED ADULT SANCTIONS

Notes: This figure summarizes the model's qualitative predictions if the age of criminal majority (ACM) is raised from T to T'. The undashed line displays crime c at each age t if the ACM was T, and the dashed line displays the analogous relationship if the ACM was set at a higher age T'. T and T' are marked as the age thresholds 17 and 18 in parentheses, in reference to the empirical analysis, which estimates the impact of raising the ACM from 17 to 18. For additional details on the model, see online Appendix A.3.

B. *Peer Effects*

This section discusses the predictions of a model in which the return to criminal activity increases as one's peers commit more crime. This discussion is motivated by research that shows that peer influences within schools and neighborhoods affect juvenile crime, including Kline (2011); Drago and Galbiati (2012); Patacchini and Zenou (2012); Billings, Deming, and Ross (2019) and Billings and Schnepel (2020). The predictions of this model are straightforward and are very similar to those reflected in Figure 4. When the ACM is raised from T to T', three groups are expected to increase criminal activity: age groups between T and T', age groups below T, and age groups above T'.

The first group predicted to respond consists of individuals aged between T and T'. This group was subject to higher, adult-level sanctions before the policy change, but is subject to lower, juvenile-level sanctions after the policy change T'. This causes an increase in criminal activity through two separate channels. First, the immediate return to criminal activity is higher, because expected sanctions (if caught) in the current period are lower; second, this increase in crime raises the return to committing crime for their peers aged between T and T'. However, this group is also subject to a simultaneous increase in underrecording since they were treated as adults previously, but are now treated as juveniles by the criminal justice system. Therefore, we may not observe an increase in official crime statistics for age groups between T and T', even if the decrease in sanctions leads to an increase in crime.

The return to criminal activity for those below T and above T' also increases as their peers between T and T' increase their level of criminal activity. This predicts an *immediate* increase in crime by both those below T and those above T'. Crucially, these age groups are treated as juveniles and adults respectively, both before and after the ACM change, and the predicted increase in crime is unlikely to be offset by changes in underrecording.

IV. Empirical Strategy

This section describes the difference-in-difference-in-differences (DDD) framework used to identify the impact of the ACM increase on adolescent offending. I also describe an event study specification, which is used to examine year-by-year effects of the ACM change, and rule out differential pre-trends.[26]

Throughout this section, I refer to the group of states that raised the ACM during the study period 2005–2016 as the treatment states, and the group with no such change as the comparison states. The DDD technique compares those who are predicted to be affected by the ACM increase (adolescents, aged 21 and below) with individuals that are not (older adults, aged 22 and above) across the two groups of states, before and after the ACM change. Examining effects on age groups below and above 17 (the original ACM) allows us to test predictions based on the models discussed in the previous section. The identifying assumption is that in the absence of the ACM change, the difference in outcomes between adolescents (ages $\leq$ 21) and older adults (ages $\geq$ 22) in treatment states would have evolved similarly to that in comparison states.

Central Specification.—Online Appendix Table A.1 specifies when the ACM was modified in each of the treatment states. This information is used to estimate the following equation:

$$C_{alst} = \beta_1 ADOL_a \times POST_{st} + \gamma_{as} + \gamma_{st} + \gamma_{at} + \varepsilon_{alst}. \tag{1}$$

C_{alst} is a measure of the crime rate among age group a in location l in state s in month t. Age-state interactions γ_{as} allow for permanent differences across age groups in different states. Age-time interactions γ_{at} control flexibly for national trends that may affect one age group more than another. State-time interactions γ_{st} control flexibly for factors changing at the state-time level that could affect crime rates.

The coefficient of interest is β_1, the DDD estimate of the effect of an ACM increase on adolescent offending. $ADOL_a$ is an indicator variable that equals one

[26] Recent work on difference-in-differences(-in-differences) with staggered treatment dates has identified potential issues with conventional two-way (three-way) fixed effects estimators. Specifically, these estimators make implicit assumptions that restrict treatment effect heterogeneity, which can produce unreliable estimates of average treatment effects when effects are heterogeneous across groups and time periods. Section VA shows that the main results do not change meaningfully when re-estimated using Borusyak, Jaravel, and Spiess' (2021) robust and efficient estimator, which does not restrict the heterogeneity of treatment effects. This estimator is constructed in three steps—(i) group and time period fixed effects are fitted by regression on untreated observations only; (ii) these are used to impute untreated potential outcomes, and obtain an estimated treatment effect for each treated observation; (iii) a weighted average of these treatment effect estimates is taken, corresponding to the estimation target.

for age groups 21 years old and under. This allows us to capture the ACM's effects not just on 17-year-olds, but on four age groups below and above 17 (the original ACM), consistent with the predictions of several models of criminal behavior.[27] $POST_{st}$ is an indicator variable that equals one if an ACM change in state s is in effect at time t.[28] Since treatment varies at the age level within each state, standard errors ε_{alst} are clustered at the age-state level.

Event Study Specification.—In order to examine the year-by-year impact of the ACM change, I use the following event study specification:

$$C_{alst} = \sum_{i \geq -n} \beta_i ADOL_a \times POST^i_{st} + \gamma_{as} + \gamma_{st} + \gamma_{at} + \varepsilon_{alst}. \tag{2}$$

C_{alst} is a measure of the crime rate among age group a in location l in state s in month t. $POST^i_{st}$ are indicator variables that equal 1 if the ACM was increased in state s exactly i years before period t. For instance, Connecticut raised its ACM from 17 to 18 on July 1, 2012, so the $POST^1$ dummy equals 1 for Connecticut during July 2012–June 2013, the $POST^2$ dummy equals 1 for Connecticut for the period July 2013–June 2014, and so on. i may take on negative values, which allows us to test for differences prior to the policy's implementation. Regressions continue to control for age-state, state-time, and age-time interactions. Standard errors are clustered at the age-state level to adjust for serial correlation.[29]

Regression Sample.—UCR arrest data, which contain monthly counts of arrests by age, sex, and offense within an agency's jurisdiction, form the basis for the main regression sample.[30] Prior research has noted the existence of errors in the UCR data and the need for thorough cleaning before use. To minimize data quality concerns:

(i) The sample is restricted to 487 law enforcement agencies that
   (a) report positive population covered every month;
   (b) submit data for every month during 2005-2016 (i.e., a balanced panel).

(ii) Rape offenses are excluded, since the UCR definition of rape offenses changed in 2013.

[27] Section IIIC shows that the estimated increase in juvenile crime is robust to redefining the comparison group to include only younger or older cohorts.

[28] Rhode Island *lowered* its ACM from 18 to 17 for the period July–November 2007. $POST_{s=RhodeIsland,t}$ takes on the value -1 for these five months, which ensures that β_1 can be interpreted as the impact of an increase in the ACM.

[29] Since Rhode Island only changed its ACM for four months before reversing it back, it is included in the comparison group for the event study regressions.

[30] UCR data, which included law enforcement agencies representing over 90 percent of the US population in 2015, is used instead of NIBRS data, because the latter's coverage is very low. As of 2015, only 30 percent of the US population was covered by NIBRS. Restricting attention to just the six Northeastern states does not meaningfully improve the coverage rate—e.g., in 2013, NIBRS covered only 34 percent of residents in the six states. Sources: https://ucr.fbi.gov/nibrs/2015/tables/data-declarations/dd_number_of_agencies_and_population_covered_by_population_group_2015_final.pdf; https://ucr.fbi.gov/nibrs/2013/resources/nibrs-participation-by-state, https://www.census.gov/data/ tables/time-series/demo/popest/2010s-state-total.html.

Table 2—Impact of an Increase in the Age of Criminal Majority

Age group	Adolescents	Juveniles		Young adults
	13–21	13–16	17	18–21
Arrest rates				
DDD estimate	0.222	0.412	0.067	0.119
	(0.067)	(0.070)	(0.284)	(0.077)
Mean	1.900	1.294	2.524	2.198
Violent crime index				
DDD estimate	0.006	0.011	0.008	0.002
	(0.004)	(0.005)	(0.009)	(0.005)
Mean	−0.021	−0.020	−0.024	−0.021
Property crime index				
DDD estimate	0.038	0.052	0.029	0.030
	(0.005)	(0.006)	(0.015)	(0.006)
Mean	−0.038	−0.042	−0.031	−0.036
Observations	1,332,432	981,792	841,536	1,051,920
Clusters	114	84	72	90

Notes: This table displays DDD estimates of the impact of an increase in the age of criminal majority from 17 to 18. The dependent variable in the first panel is the age-specific monthly arrest rate, defined as the number of arrests by age per 100,000 residents; each crime index is the average of the z-scores of its components, which are calculated by subtracting the comparison group mean and dividing by the comparison group standard deviation. The sample includes a balanced panel of 487 law enforcement agencies in the six contiguous states of Connecticut, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont, each of which introduced legislation to raise the age of criminal majority during the study period. Standard errors are clustered at the age-state level.

Source: 2005–2016 data from the FBI Uniform Crime Reports

(iii) The data is manually checked for data entry errors (e.g., 99 or 999 instead of a missing value code).

(iv) As an additional robustness check, I follow Evans and Owens (2007) and
   (a) regress total index crime in each agency on a quadratic time trend;
   (b) drop agency-years where the actual and predicted values differ by more than 50 percent in absolute value (6 percent of the sample);
   (c) re-estimate the main results; online Appendix Table A.2 shows that the revised estimates are not meaningfully different from those in Table 2.

UCR data groups arrests for those aged 13–14, reports separate counts for each age between 15 and 24, and groups arrests for those aged 25–29, 30–34, … 60–64. For each age group, the arrest rate is computed by dividing the number of arrests by population and the number of ages that the count represents. To minimize underreporting concerns to the extent possible, attention is restricted to arrests for offenses that are regarded as serious and tend to be reported reliably—i.e., the core Index 1 violent (murder, robbery, and aggravated assault) and property (burglary, larceny, and motor vehicle theft) crimes.[31] Arrest rates for those aged 13–21 serve as treatment

[31] Rape is excluded because the FBI definition for this offense changed in 2013, which is in middle of the study period.

groups, while those for 22–64 serve as comparison groups. The main results are only strengthened by using younger ages (i.e., those aged 22–29) as comparison groups, as discussed in Section VC.

Overall arrest rates may be heavily driven by frequently occurring offenses, such as theft, and less so by offenses with high social costs that occur less frequently, such as robbery or assault. To address this concern, two additional dependent variables are constructed: a violent crime index and a property crime index. Each index is defined as the equally weighted average of z-scores of its components, and z-scores are calculated by subtracting the comparison group mean and dividing by the comparison group standard deviation.[32]

V. Results

This section shows that delaying the threat of adult sanctions can lead to an increase in adolescent offending. When the ACM is increased from 17 to 18, individuals aged 17 and under in the study sample increase criminal activity. This increase is concentrated among 13–16-year-olds, consistent with the argument that a simultaneous increase in underrecording of crime by 17-year-olds would obfuscate any effects on this age group. Additionally, the ACM increase leads to a delayed increase in crime by 18-year-olds, consistent with the predictions of a model in which individuals accumulate criminal capital. The section concludes by presenting a range of robustness checks, and discussing some costs and benefits of raising the ACM.

A. *Arrest Rates*

To examine the year-by-year impact of increasing the ACM from 17 to 18 on adolescent arrest rates, Figure 5 plots point estimates and confidence intervals five years before and five years after the policy's enactment. As expected from quasi-random implementation dates, there is no statistical difference between the treatment and comparison group coefficients prior to the ACM increase. Additionally, there is a clear increase in arrest rates starting in the second year of the ACM change, with the effect size increasing over time.[33]

To dig into the delayed increase in arrest rates, panel B of Figure 5 examines effects separately for those aged 13–16, 17, and 18–21; confidence intervals are suppressed to enable clear exposition. A visible increase is observed for 13–16-year-olds starting in the first year of the policy's implementation. For 17-year-olds, there is a substantive drop in the first year of the policy's implementation, in line with the argument that underrecording of crime by this age group is expected to increase. Finally, for 18–21-year-olds, we observe a delayed increase in arrest rates, which only catches up in magnitude with the increase for 13–16-year-olds several years after the policy's implementation, despite mean arrest rates being substantively

[32] Each component of the index has mean 0 and standard deviation 1 for the comparison group; the components of the violent crime index are arrest rates for homicide, robbery, and aggravated assault; the components of the property crime index are arrest rates for burglary, larceny, and motor vehicle theft.

[33] Online Appendix Figure A.6 displays event study graphs based on Borusyak, Jaravel, and Spiess' (2021) robust and efficient estimator; the pattern of results remains unchanged.

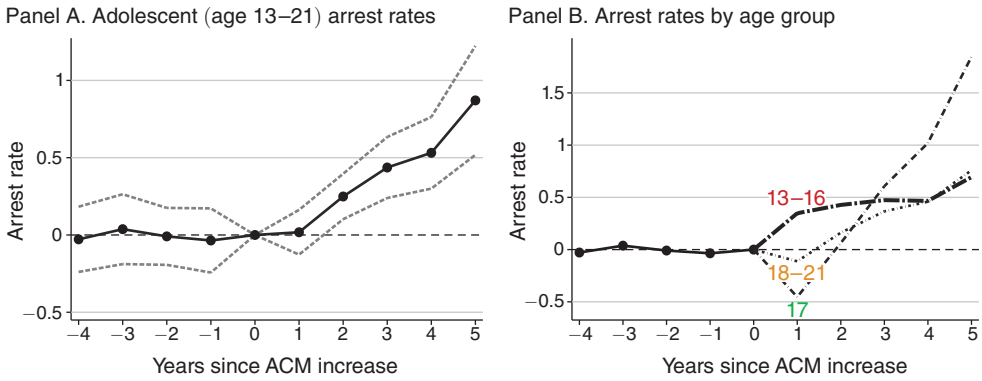

FIGURE 5. IMPACT OF AN INCREASE IN THE AGE OF CRIMINAL MAJORITY

Notes: This figure displays point estimates (and in panel A, 95 percent confidence intervals) of the year-by-year impact of an increase in the age of criminal majority from 17 to 18 on arrest rates for all adolescents (i.e., those aged 13 to 21) in panel B, and separately for those aged 13–16, 17, and 18–21 in panel B.

Source: 2005–2016 data from the FBI Uniform Crime Reports

higher for 18–21-year-olds. It is worth noting that point estimates in the first year after the ACM increase are unadulterated by the movement of individuals with longer arrest histories into older age groups, but the same cannot be said of point estimates in years 2, 3, 4, and 5. As a result, point estimates for these later years may reflect increasing effect sizes over time, and/or the movement of individuals with longer arrest histories into older age groups.

Table 2 displays corresponding point estimates and standard errors. The first panel shows that monthly arrest rates increase significantly by around 12 percent of the mean for each age group in the range 13–21.[34] The last three columns examine effects separately by age group: this effect is largely driven by 13–16-year-olds, for which we observe a statistically significant increase in arrest rates on the order of about 30 percent of the mean; effects for 17- and 18–21-year-olds are meaningfully smaller in magnitude, and not statistically significant at conventional levels.[35]

The second and third panels show the estimated impact on the violent crime index and property crime index, respectively.[36] There is a statistically significant

[34] The first panel of online Appendix Table A.3 displays analogous estimates based on Borusyak, Jaravel, and Spiess' (2021) robust and efficient estimator; point estimates are larger and remain statistically significant.

[35] We can use a back-of-the-envelope calculation to quantify the change in arresting behavior when the ACM increases. Specifically, if we expect similar crime increases for 13–16- and 17-year-olds, we would expect arrest rates to increase by 32 percent for 17-year-olds as well, not just 3 percent (as shown in Table 2). $(1 + \%$ increase in 17-year-old arrest rates$) = (1 + \%$ increase in offending$) \times (1 - \%$ decrease in recording$) \Rightarrow \%$ decrease in recording $= 1 - 1.03/1.32 = 0.22$, i.e., the rate at which police arrested 17-year-olds may have fallen by as much as 22 percent. This estimate is large, but supported by Figure 2, which uses NIBRS data—17-year-olds account for 8.2 percent of arrests in states that set the ACM at 17 but only 5.5 percent in states that set the ACM at 18. This amounts to a *reduction* of 33 percent despite 17-year-olds facing juvenile (less severe) sanctions in the latter set of states.

[36] Arrest rates are heavily influenced by frequently occurring offenses (e.g., theft) and less so by offenses with high social costs (e.g., robbery). Using these indices circumvents this issue, as each is defined as the equally weighted average of z-scores of its component offenses. z-scores are calculated by subtracting the comparison group mean and dividing by the comparison group standard deviation.

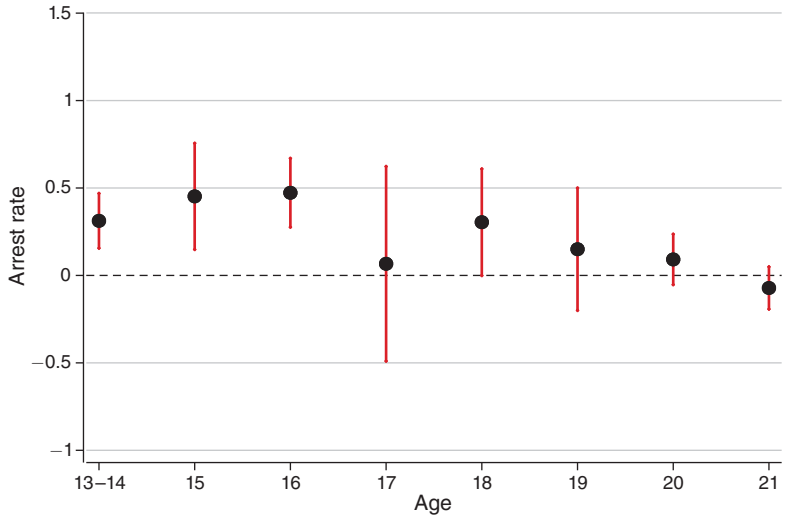

FIGURE 6. IMPACT OF AN INCREASE IN THE AGE OF CRIMINAL MAJORITY AGE-SPECIFIC ESTIMATES

Notes: This figure displays DDD estimates and 95 percent confidence intervals of the impact of an increase in the age of criminal majority from 17 to 18 on age-specific arrest rates.

Source: 2005–2016 data from the FBI Uniform Crime Reports

increase in both the violent and property crime indices for 13–16-year-olds, and just the property crime index for 17- and 18–21-year-olds.[37] Point estimates are smaller for the violent crime index across age groups, consistent with (i) higher expected punishments for these crimes, so that there are fewer people on the margin of committing these offenses, and (ii) the ACM increase being less likely to lower the probability of facing adult sanctions for these offenses.[38] Taken together, these estimates indicate that those above the ACM react to its shift over time, but that the largest effects are observed for those *below* the threshold, in terms of both overall arrest rates as well as more socially costly violent crimes.

B. *Effects By Age, Sex, and Race*

This section examines whether some demographic groups are more responsive to the increase in the ACM than others.

Age.—Figure 6 displays the estimated impact on arrest rates for each age group in the age range 13–21. Thirteen- and 14-year-olds are clubbed together because the Uniform Crime Reports only include collective data for 13- and 14-year-old

[37] The last two panels of online Appendix Table A.3 display analogous estimates based on Borusyak, Jaravel, and Spiess' (2021) robust and efficient estimator; point estimates are larger and remain statistically significant.

[38] National statistics from 2005 show that person offenses (e.g., assault) accounted for 4 percent of juvenile arrests but 51 percent of cases transferred to adult court; property offenses (e.g., larceny) accounted for 20 percent of juvenile arrests but 27 percent of cases transferred to adult court (Puzzanchera and Sickmund 2005; table 43: https://ucr.fbi.gov/crime-in-the-u.s/2005/crime-in-the-US-2005/.

Table 3—Impact of an Increase in the Age of Criminal Majority on Juvenile Arrest Rates by Age, Sex, and Race

	Male 13–14	Male 15	Male 16	Male 17
DDD estimate	0.290	0.430	0.452	0.167
	(0.071)	(0.104)	(0.092)	(0.131)
Mean	0.482	0.910	1.216	1.611
	Female 13–14	Female 15	Female 16	Female 17
DDD estimate	0.212	0.211	0.210	0.089
	(0.064)	(0.077)	(0.071)	(0.145)
Mean	0.231	0.423	0.619	0.913
	White 0–17	Black 0–17	Other 0–17	
DDD estimate	1.082	0.177	0.140	
	(0.488)	(0.186)	(0.057)	
Mean	5.748	1.573	0.085	
Observations	841,536	841,536	841,536	841,536
Clusters	72	72	72	72

Notes: This table displays DDD estimates of the impact of an increase in the age of criminal majority from 17 to 18 on age-, sex-, and race-specific monthly arrest rates. The sample includes a balanced panel of 487 law enforcement agencies in the six contiguous states of Connecticut, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont, each of which introduced legislation to raise the age of criminal majority during the study period. Standard errors are clustered at the age-state level. The UCR data does not contain age specific arrests by race, only the number of arrests under 18 separated by race; Hispanic arrestees are not reported separately and may belong to any of the race categories.

Source: 2005–2016 data from the FBI Uniform Crime Reports

arrestees. Arrest rates increase significantly for 13–16-year-olds as well as for 18-year-olds.

The smaller (and noisier) estimate for 17-year-olds is particularly conspicuous, consistent with the argument that 17-year-olds are exposed simultaneously to lower sanctions as well as other changes by dint of now being below the ACM, including but not limited to an increase in underrecording of crime. The latter effect appears to offset the effect of lower sanctions, which can be observed much more clearly for those aged 13–16 and 18. It is worth noting, however, that this effect reflects an average across post-treatment years; Figure 5 and online Appendix Figure A.6 show, for instance, that the effect on 17-year-olds is likely to grow over time.

Sex.—Table 3 splits the data even further into arrests by age and sex. The first panel shows the estimated impact on male arrest rates by age, while the second panel shows analogous estimates for female arrest rates. Both panels show that there are large, statistically significant increases in arrest rates for all age groups *except* 17, consistent with the argument that 17-year-olds may be exposed to several other changes following the ACM increase, including a reduction in the probability of being arrested for any given crime.

Race.—The Uniform Crime Reports also include data on arrests for individuals aged 17 and under by race (though finer data on age- and race-specific arrests are not included). The last panel of Table 3 shows that the estimates presented above are

largely driven by an increase in the arrest rate for White youth (an increase equivalent to 19 percent of the mean), while the response among Black youth is smaller in magnitude and statistically indistinguishable from zero.[39] This finding is consistent with effective treatment differing across race groups; if Black youth are more likely to be charged in adult courts (Juszkiewicz 2009; Zane, Cochran, and Mears 2022) or face harsher sanctions within the juvenile justice system (Eren and Mocan 2018), raising the ACM may change their incentives less than those of other youth.[40] This finding is also supported by Lovett and Xue (2018), who use administrative data from California to show that White and Asian youth respond to the sanction hike at the ACM, while Black youth do not.

C. *Robustness Checks*

This section shows that the estimated increase in arrest rates for 13–16-year-olds survives a number of tests such as using alternative age groups as comparison groups, accounting for geographical spillovers, and extending the study sample to include all states in the United States.

Alternative Age Groups as Comparison Groups.—Heretofore, the analysis has used individuals aged 22 and above as comparison groups. Figure 7 shows that the estimated increase in juvenile arrests is not driven by this choice. Estimates remain positive and statistically significant when the comparison group is redefined to only include younger (below 30) or older (30 and above) age groups. The first group of estimates corresponds to the six Northeastern states included in the study sample (labeled 6 NE)—estimates remain similar in magnitude and statistically significant across the three specifications.

Geographical Spillovers.—One possible mechanism behind the increase in juvenile arrests may be that crime is simply shifting over to treatment states from comparison states, with no increase in overall crime. To test this hypothesis, I reestimate the DDD coefficients after excluding jurisdictions that border treatment and comparison states.[41] These are the second set of estimates displayed in Figure 7 (labeled 6 NE–Borders). This exercise does not meaningfully alter the estimates, irrespective of which age groups we use as comparisons.

Estimates Not Driven by One Specific State.—Given the limited number of states in the sample, an additional concern may be that trends in a single state are driving the overall findings. To show that this is not the case, the next six sets of estimates in Figure 7 show how results vary when we drop a single state at a time, starting with

[39] Agencies do not separately report arrests for Hispanic arrestees, who can be included in any of the race categories.

[40] National statistics from 2005 show that Black youth accounted for 30 percent of juvenile arrests but 39 percent of cases transferred to adult court; White youth accounted for 68 percent of juvenile arrests but 58 percent of cases transferred to adult court (Puzzanchera and Sickmund2005; Table 43: https://ucr.fbi.gov/crime-in-the-u.s/2005/crime-in-the-US-2005/).

[41] Online Appendix Table A.4 displays the list of law enforcement agencies that are excluded from the analysis.

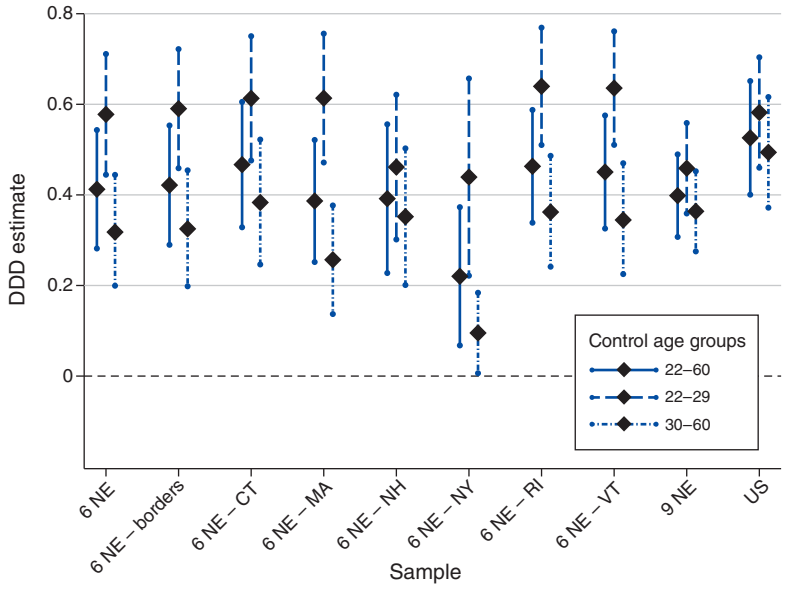

FIGURE 7. ROBUSTNESS CHECKS

Notes: This figure displays DDD estimates and 95 percent confidence intervals of the impact of an increase in the age of criminal majority from 17 to 18 on 13–16-year-old arrest rates for different subsets of the data.

Source: 2005–2016 data from the FBI Uniform Crime Reports

Connecticut (labeled 6 NE–CT) and ending with Vermont (labeled 6 NE–VT). Point estimates remain positive and statistically significant across these specifications.

Accompanying Juvenile Justice Reforms.—A natural concern with studies that exploit ACM changes is that they may be accompanied by other juvenile justice reforms, which in turn may be driving the estimates. This concern is partly assuaged by the fact that we observe a delayed reaction to the ACM by 18-year-olds, who are treated as adults both before and after the ACM change, and would be unaffected by other reforms that explicitly target juveniles.

Among the treatment states, Connecticut implemented a range of juvenile justice reforms such as reducing in-school arrests in 2009 and discontinuing the detention of juveniles for noncriminal cases in 2007. These reforms were not implemented in the same year as the ACM increase, and the event study estimates show that these policies did not have a large impact prior to the ACM change. Moreover, these policies serve to *reduce* the number of juvenile arrests, which would lead to an underestimate of the true effect of the ACM increase. In Figure 7, the third set of estimates (labeled 6 NE–CT) show how the estimates change when we exclude Connecticut from the study sample: the point estimate is slightly larger, consistent with the fact that accompanying reforms would reduce juvenile arrests.

Including Additional States.—As a final check, I show that the main results are robust to widening the sample to include additional states. The second-to-last set of

estimates (labeled 9 NE) displayed in Figure 7 use all nine states in the Northeast, even those that did not introduce legislation to raise the ACM during the study period (Maine, New Jersey, and Pennsylvania). The last set of estimates use data from all states that report crime via the UCR, which now includes two additional treatment states: Illinois (post-2009) and Mississippi (post-June 2011). Both sets of estimates remain large and statistically significant, irrespective of which age groups are used as comparisons.

D. *Some Benefits and Costs of Raising the Age of Criminal Majority*

This section uses a back-of-the-envelope calculation to estimate the social costs of raising the ACM. This is a partial estimation exercise, as it is based on multiple assumptions and focuses only on two sources of social cost increases due to the ACM change: the increase in criminal offending by 13–16-year-olds, and the costs of transferring 17-year-olds to more expensive juvenile facilities. For every 17-year-old that is expected to be processed through the juvenile justice system, the ACM increase leads to an increase of $86,325 in social costs. This section concludes by discussing the need for more research to understand whether these costs will be offset by the benefits of raising the ACM and some tools that local jurisdictions can employ to anticipate and offset these crime increases.

The first source of social costs is the increase in criminal offending by age groups below 17. The second column of online Appendix Table A.5 displays the estimated impact of the ACM increase on the social costs of 13–16-year-old *arrests*—an increase of $2,222 per month per 100,000 residents for each age group within the 13–16 age group.[42] To predict the annual increase in *offending* by 13–16-year-olds, this cost estimate is adjusted by the 2015 arrest-to-offense ratio of 0.213 for the six Index offenses in the Uniform Crime Reports.[43] Evaluated at the mean agency population of 30,051, the crime increase among 13–16-year-olds is estimated to lead to an annual increase of $150,453 in social costs.

The second source of social costs is the transfer of 17-year-olds to juvenile facilities. Juvenile incarceration costs in the treatment states average $544 per day (Justice Policy Institute 2014), while the equivalent estimate for adult incarceration is $198 (Vera Institute of Justice 2017).[44] Online Appendix Table A.6 summarizes the steps implemented to estimate of the number of such transfers in the average agency within the study sample. The first column shows the monthly arrest rate for 17-year-olds

[42] I adjust Autor, Palmer, and Pathak's (2017) estimates of the cost of crime accrued by victims and the criminal justice system—based on Cohen and Piquero (2009)—for inflation; estimates of offenders' productivity losses are excluded since individuals below the age of 17 are unlikely to be a part of the formal labor force. This exercise results in cost estimates of $67,896 and $3,626 for violent and property crimes, respectively, in 2015 US$.

[43] Sources: https://ucr.fbi.gov/crime-in-the-u.s/2015/crime-in-the-US-2015/tables/table-1; https://ucr.fbi.gov/crime-in-the-u.s/2015/crime-in-the-US-2015/tables/table-29. This ratio does not include offenses that are not reported to the police and is, therefore, an underestimate of the total increase in offending. This method will also underestimate the increase in offending if juveniles are arrested at lower rates than adults. I do not account for the fact that 13–16-year-olds who offend and are incarcerated at higher rates may be more likely to recidivate in the future (Aizer and Doyle 2015).

[44] Since New Hampshire is not included in the Vera Institute of Justice (2017) report, I use the Vera Institute of Justice (2012) estimates assuming that its costs grew at the same rate as Connecticut and Rhode Island, which provided information in both surveys. Estimates are in 2015 USD.

in the study sample, and the second displays the ratio of offense-specific juvenile arrests to juvenile court cases in 2015, based on the Uniform Crime Reports and the National Center for Juvenile Justice (2015). I combine this with offense-specific incarceration rates for 17-year-olds[45] along with expected sentence lengths specific to each offense category (person, property, drug and public order), relying on data from the Office of Juvenile Justice and Delinquency Prevention and the National Center for Juvenile Justice (2015). The last two columns use the incarceration cost estimates of $544 and $198 mentioned above. The increase in costs due to the transfer of 17-year-olds from adult to juvenile facilities is $390,458.[46] Adding in the costs due to heightened offending by 13–16-year-olds gives us an estimate of $540,911 in social costs as a result of the ACM increase.

What are the benefits of raising the ACM? Proponents of raising the ACM argue that rates of recidivism are likely lower after being processed by the juvenile justice system. Recent studies show, however, that incarceration in juvenile facilities also has a large impact on recidivism (Aizer and Doyle 2015) and that adult incarceration may actually lower recidivism for marginal offenders (Entorf 2011, Pichler and Römer 2013, Loeffler and Grunwald 2015a).[47] Therefore, I do not focus on lower recidivism as the primary benefit of raising the ACM; instead, I focus on the 17-year-olds who will be diverted away from adult prisons, and will usually have the option to expunge their criminal records. My estimate of the number of 17-year-olds that will receive this benefit is displayed by offense type in the fourth column of online Appendix Table A.6, which sums up to a total of 6.266 17-year-olds.

The question for policymakers is whether costs of at least $86,325 per 17-year-old are exceeded by the potential benefits. There are many reasons why it might. One, the expungement of criminal records may boost employment and average annual real earnings by around $6,000 (Selbin, McCrary, and Epstein 2017); two, the transfer to juvenile facilities may lower the risk of assault faced by the average juvenile convict—McCollister, French, and Fang (2010) estimate victim costs alone of over $200,000 for sexual assault and $100,000 for aggravated assault;[48] three, if more 17-year-olds receive probation instead of incarceration sentences, Bayer, Hjalmarsson, and Pozen (2009) and Aizer and Doyle's (2015) findings indicate that we may see an increase in high school completion rates and a decrease in recidivism. It is worth reiterating, however, that these benefits are not well understood, and more research is needed to shed light on how large these benefits are and which subgroups benefit the most. In the meantime, policymakers implementing these reforms can

[45] Here, I make three assumptions. One, the proportion of 17-year-olds adjudicated delinquent that receive placement sentences (instead of probation sentences) does not change after the increase in the ACM. Two, the cost of probation for a 17-year-old does not change when the ACM is raised to 18. Third, the completed duration of incarceration does not depend on the ACM, an assumption supported by the findings of Fritsch, Caeti, and Hemmens (1996) and Fagan (Jan/April 1996.).

[46] If the marginal incarceration cost is around half of the average cost in both juvenile and adult facilities, as found by Owens (2009) in Maryland, the cost increase will be $195,229.

[47] Hjalmarsson (2009*b*) and Gandelman and Munyo (2020) also show that juvenile incarceration can lower post-release recidivism when compared to non-carceral sanctions.

[48] It is difficult to quantify the change in assault risk faced by adolescents across different types of facilities. For instance, Beck and Hughes (2004) document that rates of reported sexual assault are six times higher in juvenile correctional facilities than in adult facilities across the United States. This is at least partly driven by state laws specifying that all sexual acts involving youth below certain ages are nonconsensual.

be proactive in understanding that there are real costs associated with these policies, and try to offset some of these effects by expanding services such as cognitive behavioral therapy and job training to reduce juvenile criminal involvement (Heller 2014; Heller et al. 2017; Kessler et al. 2021).

VI. Conclusion

Research shows that the decision to commit crime can be influenced by one's prior involvement with crime as well as the criminal involvement of one's peers. Most research on the effects of sanctions on juvenile crime, however, has overlooked the influence of these factors.

In this paper, I show that accounting for these incentives can change how we look for and measure the impact of sanctions on juvenile crime; this approach may also help account for other changes at the threshold (e.g., differences in the rate at which juvenile and adult offending is included in official crime statistics). Using policy variation in the Northeastern states since 2005, I find that raising the ACM from 17 to 18 increased arrest rates, an increase driven primarily by 13–16-year-olds. These estimates are consistent with several models of criminal behavior, including peer effects and the accumulation of criminal skills and habits.

While this paper studies the costs of raising the ACM, more research is needed to quantify the potential benefits of raising the ACM, which are not well understood. These estimates are of particular relevance today, as states have continued to push for higher ACMs (in 2016, for instance, Vermont increased the ACM to 21 for certain offenses, while Connecticut, Massachusetts, and Illinois have each introduced legislation to do the same).[49]

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[49] For more details, see https://www.bostonglobe.com/metro/2019/07/09/crime-bill-would-redefine-juveniles-age/maHshbBT6QaaX9ooVDVidN/story.html.

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